SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I P.Kalyani(192472199) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Kalyani

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Question 1

Smart Home Automation System using Intelligent Agents

1. Understanding of Concepts

Problem Overview

A smart home automation system is designed to improve the quality of life by automatically controlling **lighting, temperature, and security**. The system continuously receives real-time information from sensors such as motion detectors, temperature sensors, light sensors, and security cameras. It must also learn user preferences over time to provide personalized services while reducing energy consumption.

The system can be implemented using different types of **Intelligent Agents**, each with different decision-making capabilities.

2. Explanation

Types of Intelligent Agents

Agent Type	Working Principle	Smart Home Example
Simple Reflex Agent	Acts only on current input using condition–action rules.	Turns ON lights immediately when motion is detected.
Model-Based Agent	Maintains an internal model of previous states and environment.	Remembers that a room is occupied even if motion stops briefly and keeps the lights ON.
Goal-Based Agent	Selects actions that help achieve predefined goals.	Adjusts room temperature to reach the user's preferred comfort level.
Utility-Based Agent	Chooses the action that provides maximum overall benefit by considering multiple factors.	Balances comfort, electricity cost, and security to make the best decision.

Working Process

Sensor Input

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▼

Environment Analysis

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▼

Intelligent Agent

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▼

Decision Making

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▼

Lighting / Temperature / Security Control

3. Application

Application in Smart Home

Lighting System

- Motion detected → Lights turn ON.
- No motion for a specified time → Lights turn OFF.
- Daylight available → Brightness automatically reduced.

Temperature Control

- Temperature sensor monitors room conditions.
- Air conditioner or heater adjusts automatically.
- Learns user preferences for different times of the day.

Security System

- Detects unknown visitors.
- Sends notifications to the homeowner.
- Activates alarms when suspicious activities are detected.

Hybrid Intelligent Agent

A combination of **Model-Based**, **Goal-Based**, and **Utility-Based Agents** provides the best performance.

Agent	Contribution
Model-Based	Learns previous user behavior.
Goal-Based	Maintains desired comfort level.
Utility-Based	Optimizes energy usage and security simultaneously.

4. Organization

Comparison of Intelligent Agents

Feature	Simple Reflex	Model-Based	Goal-Based	Utility-Based
Uses Current Sensor Data	✓	✓	✓	✓
Maintains Memory	✗	✓	✓	✓
Goal-Oriented	✗	✗	✓	✓
Optimizes Multiple Factors	✗	✗	Limited	✓
Best for Smart Home	Low	Good	Better	Best

5. Accuracy

The hybrid approach satisfies all system requirements:

- Automatic lighting control ✓
- Intelligent temperature management ✓
- Adaptive learning ✓

- Enhanced security ✓
- Energy-efficient operation ✓

6. Clarity Recommended Architecture

Sensors
(Motion, Temperature, Camera)
│
▼
Hybrid Intelligent Agent
▼ ▼ ▼
Lighting Temperature Security
│
▼
Comfort + Energy Saving + Safety

Conclusion

A Hybrid Intelligent Agent combining Model-Based, Goal-Based, and Utility-Based Agents is the most effective solution for a smart home automation system. It not only reacts to current sensor inputs but also remembers previous user behavior, works toward predefined comfort goals, and optimizes energy consumption and security. This combination ensures an intelligent, adaptive, and efficient smart home environment that enhances user comfort while minimizing energy usage and maintaining safety.

Question 2

Robot Navigation in an Unknown Maze

1. Understanding of Concepts (25%)

Problem Overview

A robot is placed inside an unknown maze and must find the exit without having prior knowledge of the environment. The robot has to explore different paths, avoid dead ends, and identify the shortest or least-cost route. Since the maze is unknown, the robot relies on **Uninformed Search Algorithms** and intelligent decision-making strategies.

2. Explanation

Uniform Cost Search (UCS)

Uniform Cost Search expands the node with the **lowest cumulative path cost**.

Working Steps

Start

↓

Insert Start Node

↓

Expand Lowest Cost Node

↓

Calculate New Costs

↓

Repeat Until Goal

Iterative Deepening Search (IDS)

IDS repeatedly performs **Depth-Limited Search** with increasing depth until the goal is found.

Working Steps

Depth = 0

↓

Depth = 1

↓

Depth = 2

↓

Continue

↓

Goal Found

Comparison

Feature	UCS	IDS
Strategy	Cost-Based	Depth-Based
Completeness	Yes	Yes
Optimal	Yes	Yes (Equal Costs)
Memory Usage	High	Low
Time Complexity	Higher	Moderate

3. Application

Robot Decision-Making

The robot continuously performs the following tasks:

- Explores unknown paths.
- Stores visited locations.
- Calculates movement costs.
- Avoids repeated exploration.
- Selects the safest route.

Relation to Intelligent Agents

Agent	Role
Simple Reflex	Moves based only on immediate surroundings.
Model-Based	Builds an internal map while exploring.
Goal-Based	Searches specifically for the maze exit.
Utility-Based	Chooses the safest and least-cost route.

Best Combination

Model-Based Agent + Uniform Cost Search

Reasons:

- Remembers explored paths.
- Prevents revisiting the same locations.
- Finds the least-cost path.
- Adapts to unknown environments.

4. Organization

Decision Flow

Robot Starts

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▼

Observe Surroundings

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▼

Update Internal Map

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▼

Uniform Cost Search

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▼

Choose Lowest Cost Path

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▼

Repeat

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▼

Reach Exit

5. Accuracy

Advantages and Disadvantages

Algorithm	Advantages	Disadvantages
UCS	Optimal solution, complete	High memory usage
IDS	Low memory usage, complete	Repeated node expansion

6. Clarity

Recommended Solution

Unknown Maze

↓

Model-Based Agent

↓

Uniform Cost Search

↓

Minimum Cost Path

↓

Maze Exit

Conclusion

For an unknown maze, the combination of a **Model-Based Intelligent Agent** and **Uniform Cost Search (UCS)** provides the most effective solution. The Model-Based Agent enables the robot to build and update an internal representation of the maze, while UCS guarantees the least-cost path to the exit. Although UCS requires more memory than IDS, its ability to find the optimal solution makes it more suitable for applications where path cost and navigation efficiency are critical. This combination results in accurate, intelligent, and reliable decision-making in unfamiliar environments.